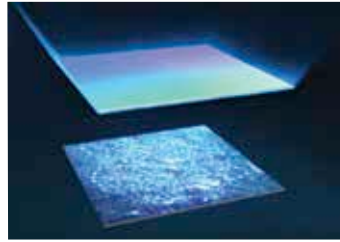


Sony announces IMX500 image sensor with AI built in

On 14 May, Sony announced the development of two new imaging sensors, the IMX500 and IMX501. What's new about these sensors is that they pack an AI chip in the sensor. Sony says the whole idea behind developing a sensor with an AI chip baked-in is to circumvent the various issues that can come with cloud-based AI processing. The Sony IMX500 is not a regular imaging sensor, but in fact, it utilises Sony's stacked sensor design. Sony had developed the ability to sandwich a DRAM layer between the optical layer of the sensor and its circuitry in order to increase the shooting buffer. This is what allowed smartphones to shoot at 960fps super slow-mo back when the feature was originally introduced. Now Sony has managed to pack an AI processing engine into this layer, instead of just having DRAM. The logic chip, as Sony calls it, includes an ISP to process the image signal, a DSP dedicated to AI signal processing and memory for the AI model.

This approach has two distinct advantages. The sensor has a resolution of 12.3 megapixels and measures 1/2.3-inch diagonally. The Sony IMX500 sounds like a sensor that could be



used in a large variety of ways. For smartphones, it could take the load off of the main SoC by running inference right on the sensor, while the higher-level AI tasks can still be conducted using the cloud.

The sensor can also be used in security cameras, where it could be programmed to, for example, identify whether the person entering the building is wearing a mask or not. The possibilities are limitless.

<https://dgit.in/sonysensor>

6-series Snapdragon processor with 5G support leaked

5G is becoming the centre play for Qualcomm this year with the chipmaker taking its 5G modems beyond the flagship level. We saw that happen with the Snapdragon 765 late last year, and now the Snapdragon 6-series will also reportedly include a 5G SoC. Details about an sm6350 SoC leaked



courtesy of noted Chinese tipster Digital Chat Station on Weibo which highlights a 5G chip inside. The chip could be a big push towards the China market where 5G connectivity is live and 5G smartphones are launching left, right and centre.

According to the tipster, the leaked 6-series Snapdragon SoC will fea-

ture an octa-core CPU design with two large cores clocked at 2.25GHz and six efficiency cores clocked at 1.8GHz. The chipset will feature the Adreno 615 GPU which also features in the Snapdragon 760. The chip is expected to launch in Q2 2020 and will sport a 5G modem. MediaTek and Huawei are also expected to announce similar mid-range 5G chipsets. The post included the name of MT6853 5G SoC and the Kirin 720 5G, which are expected to be announced soon.

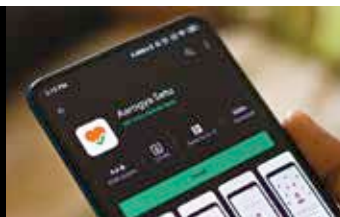
<https://dgit.in/6snap>



Social distancing with SODAR

Google has released a new tool that is aimed at helping people maintain social distancing. The tool is currently only available on Android and uses the phone's camera to visualise a white circle around the user via Augmented reality (AR). This circle has a radius of 2m.

<https://dgit.in/gosodar>



Bug bounty for Aarogya Setu

If you're a hacker or someone who knows coding in regards to apps can now help find bugs in India's own contact-tracing app Aarogya Setu. The program is open to all, participants are being asked to be responsible with disclosures, and stand to win a bounty of ₹4 lakh.

<https://dgit.in/asbb>



Hubble snaps NGC 3895

In a breath-taking image that was captured by NASA, the NGC 3895, a barred spiral galaxy can be seen clearly as a whirling mass of planets and stars. The spiral galaxy was first observed in 1790 by William Herschel, and has since been imaged many times, but never so perfectly.

<https://dgit.in/ngc3895>